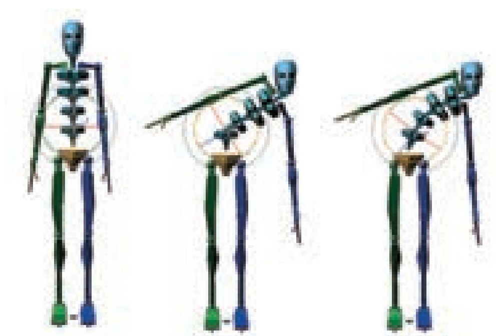
Rotating any joint in an arm or leg chain initiates forward kinematics, while moving a hand or foot initiates IK. This makes it much easier to pose the character because you don't need to switch modes.

The Center of Mass (COM) object is a tetrahedral-shaped object located near the pelvis and is used to move the Biped itself. This differs a bit from other types of skeletons, because normally the pelvis would be used for translation. A Biped, however, does not allow for the pelvis to be moved. Instead, it uses the COM object. The COM takes into account the mass of the Biped, so it is not fixed in relation to the skeleton.

Bend Links mode allows related joints of a Biped to be manipulated all at once. For example, selecting a spine joint with Bend Links activated allows rotation of one spine joint to affect all spine joints. The Track Selection rollout gives you tools for manipulating the COM as well as selecting mirror and symmetrical joints on the Biped.

- **Body Horizontal**
  Allows you to move and animate the centre of mass horizontally
- **Body Vertical**
  Allows you to move and animate the centre of mass vertically
- **Body Rotation**
  Allows you to rotate the centre of mass
- **Lock COM**
  Allows you to turn on Body Horizontal, Vertical, and Rotation all at once
- **Symmetrical**
  Selects symmetrical joints, so selecting the right hand will also select the left
- **Mirror**
  Selects the same joint (or joints) on the opposite side of the body



Rotating a spine joint without Bend Links (left). When Bend Links is enabled, all the joints rotate evenly (right).

8.3.1 Key framing Tools

The Key framing Tools rollout provides tools to clear and mirror animation on a Biped or selected parts, separate out tracks for key framing, and anchor joints to a point in space. By default, 3ds Max stores keys for an entire limb rather than the individual joints. Generally, this makes animation easier,